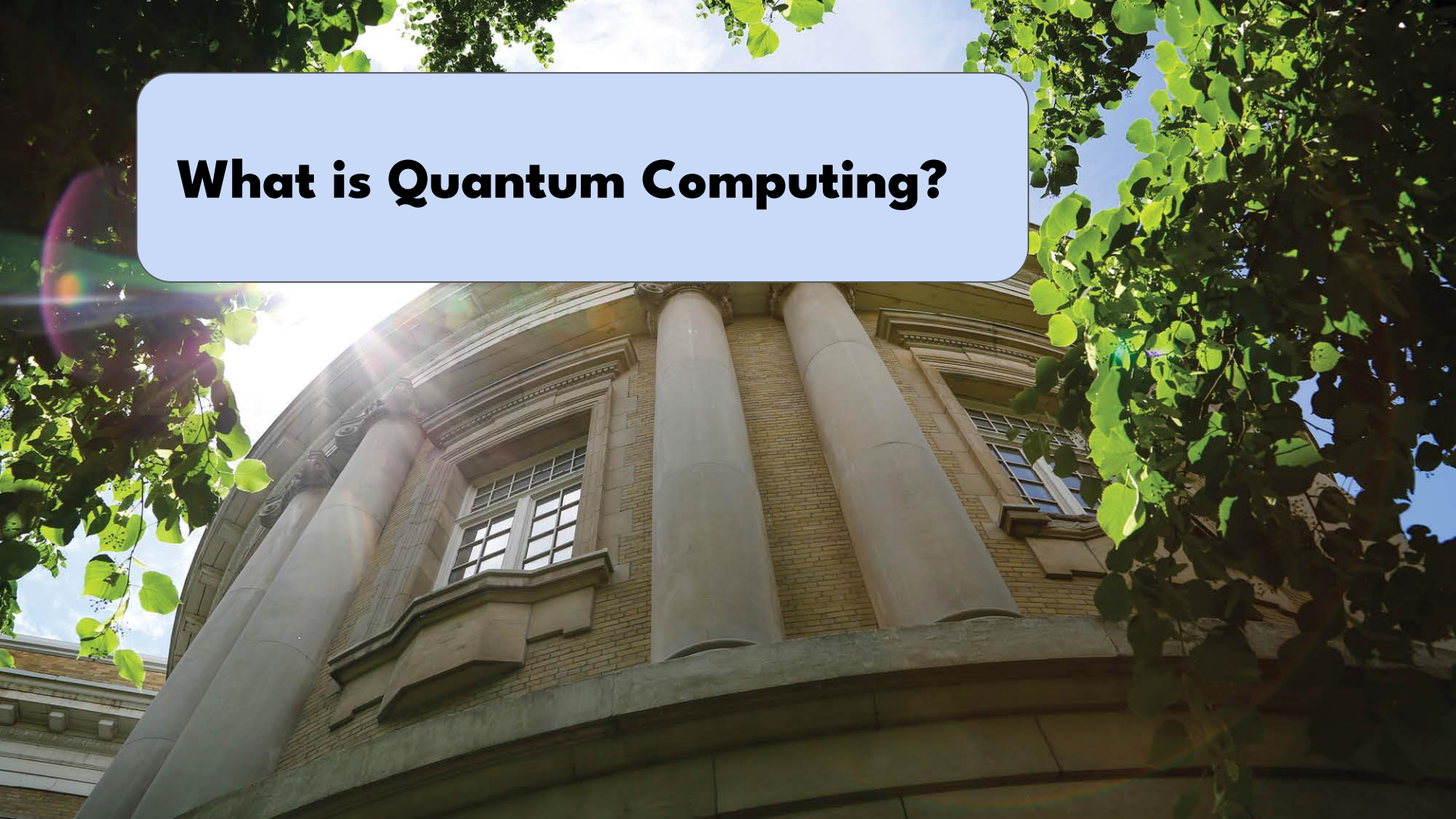
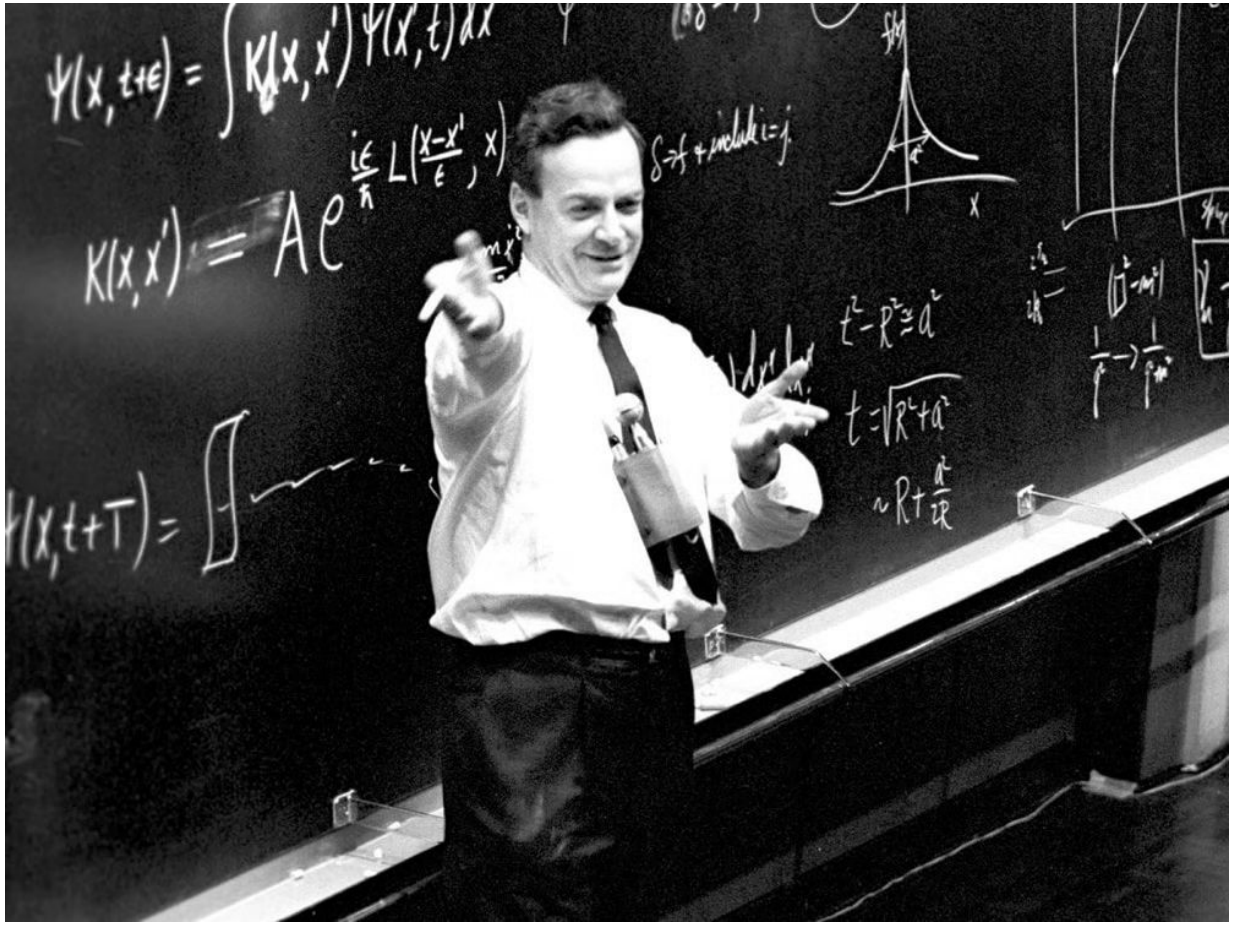
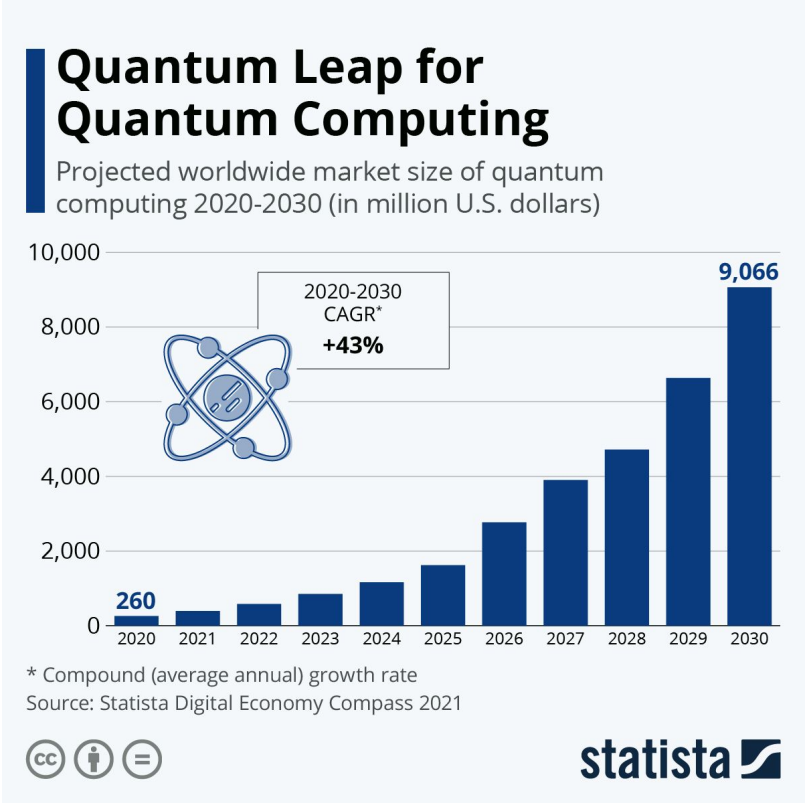


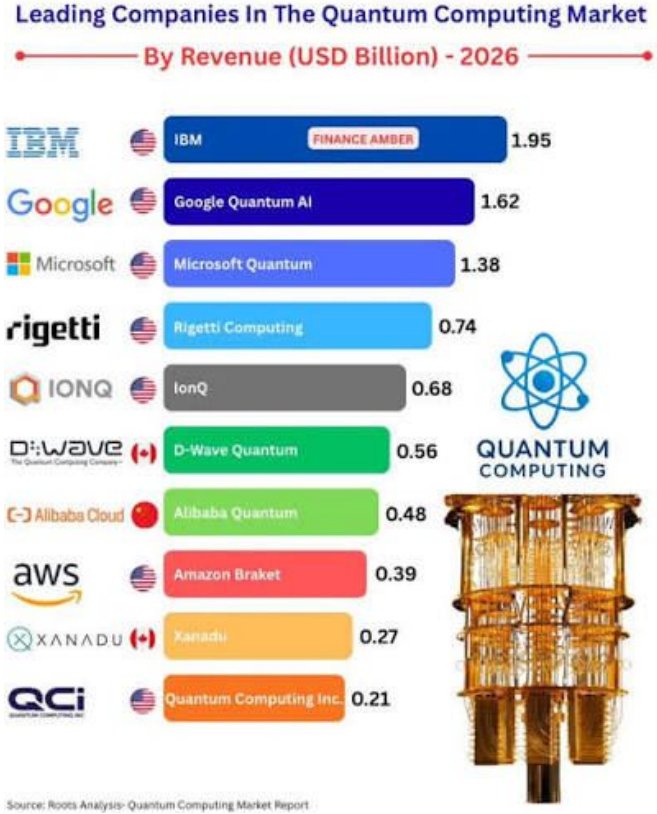
# What is Quantum Computing?




*'If you think you understand quantum mechanics, you don't understand quantum mechanics'* - Richard Feynman

# Why Quantum?

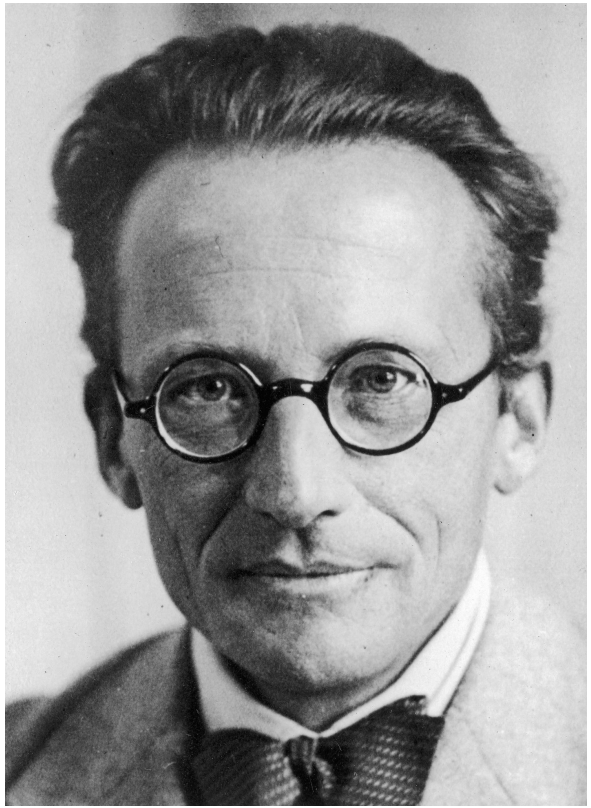

Quantum Leap for Quantum Computing
Projected worldwide market size of quantum computing 2020-2030 (in million U.S. dollars)
10,000
8,000
6,000
4,000
2,000
0
2020-2030 CAGR*
+43%
260
9,066
2020
2021
2022
2023
2024
2025
2026
2027
2028
2029
2030
* Compound (average annual) growth rate
Source: Statista Digital Economy Compass 2021
statista


Leading Companies In The Quantum Computing Market
By Revenue (USD Billion) - 2026
IBM
IBM
FINANCE AMBER
1.95
Google
Google Quantum AI
1.62
Microsoft
Microsoft Quantum
1.38
rigetti
Rigetti Computing
0.74
IONQ
IonQ
0.68
D-Wave
D-Wave Quantum
0.56
Alibaba Cloud
Alibaba Quantum
0.48
aws
Amazon Braket
0.39
XANADU
Xanadu
0.27
QCi
Quantum Computing Inc.
0.21
QUANTUM COMPUTING
Source: Roots Analysis- Quantum Computing Market Report

# In classical computers…



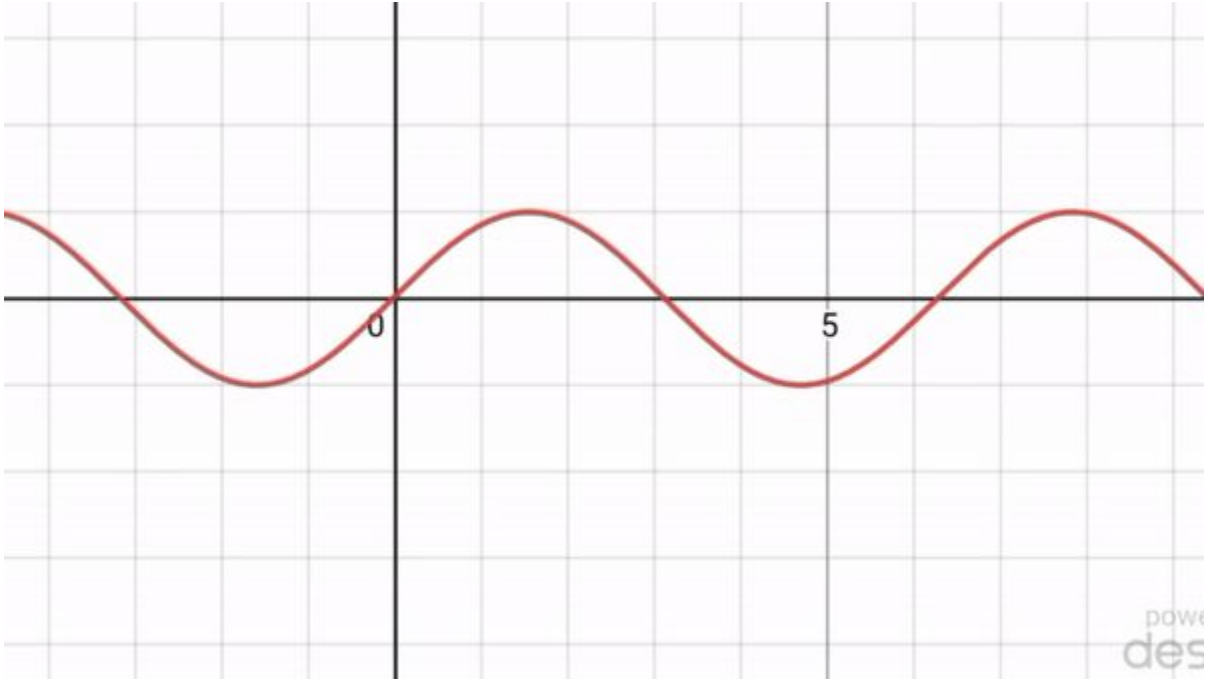
# Quantum Mechanics in 1 minute!



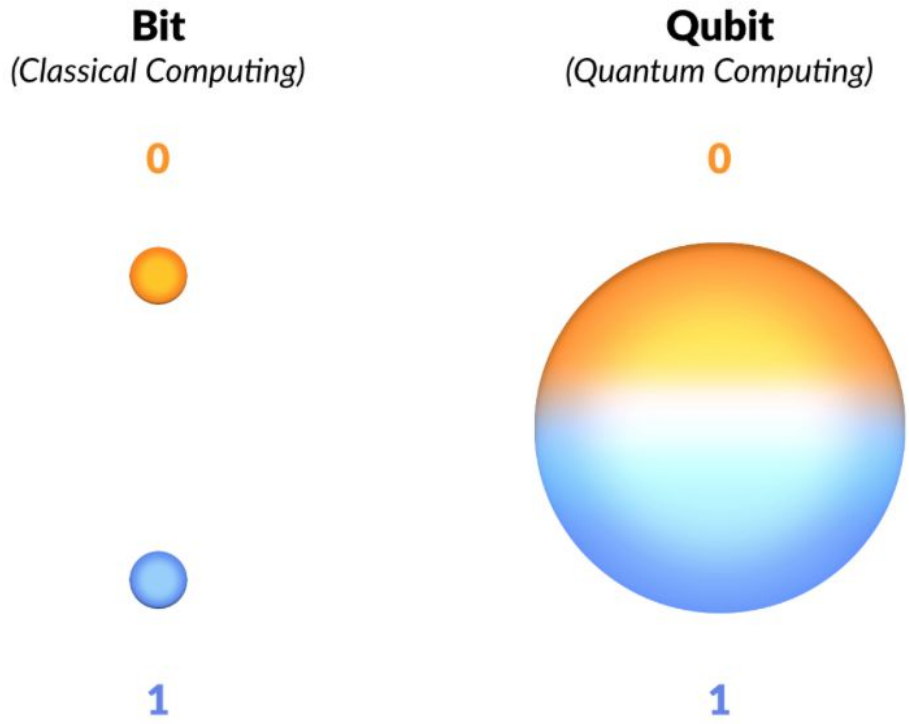
$$i\hbar\frac{\partial}{\partial t}\left|\Psi\right\rangle = \hat{H}\left|\Psi\right\rangle$$

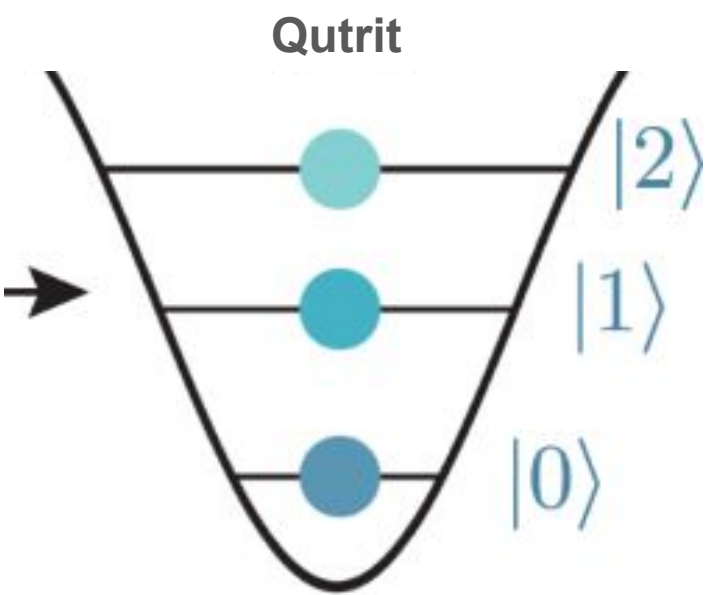
*Schrodinger's Equation*

# In Quantum Computers…



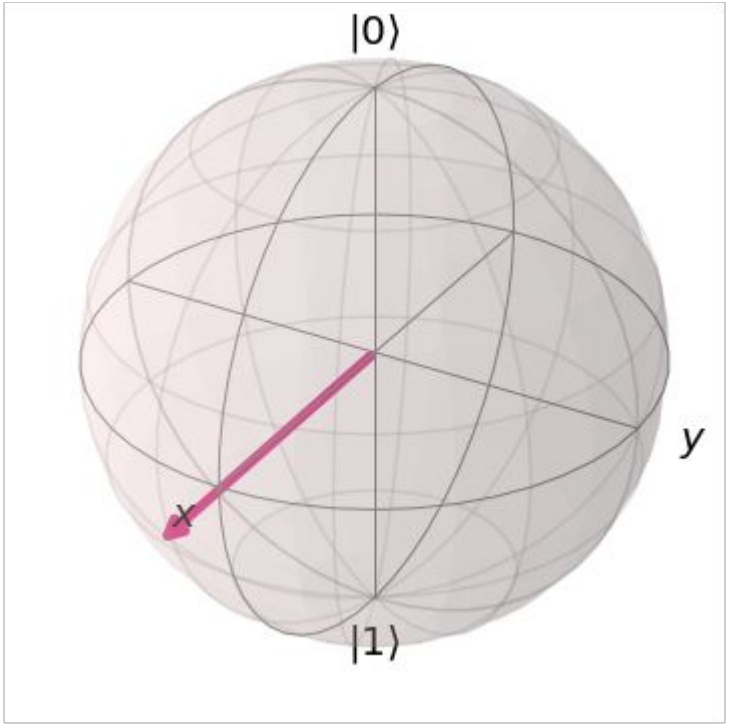
# Qubits… and beyond!



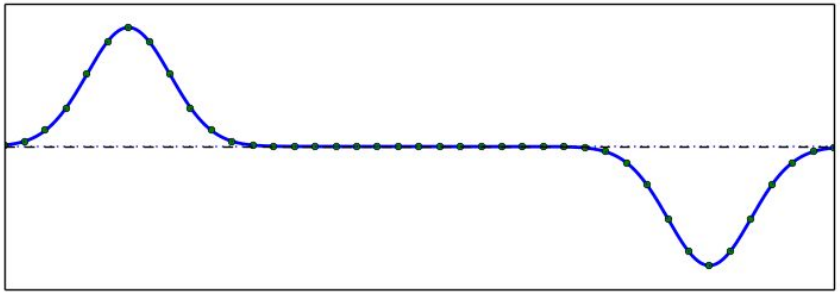



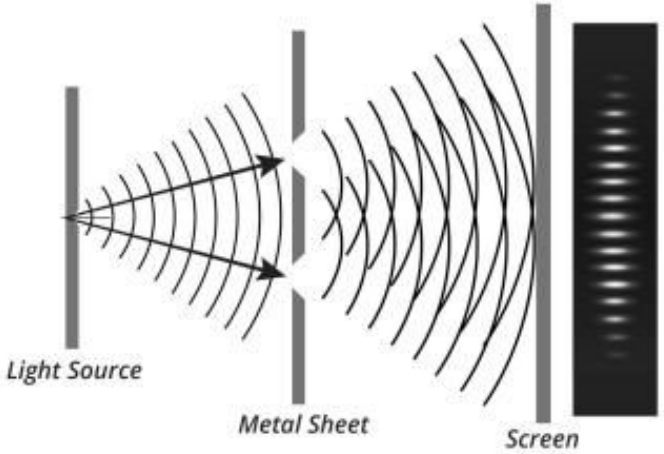
# Principle #1: Superposition





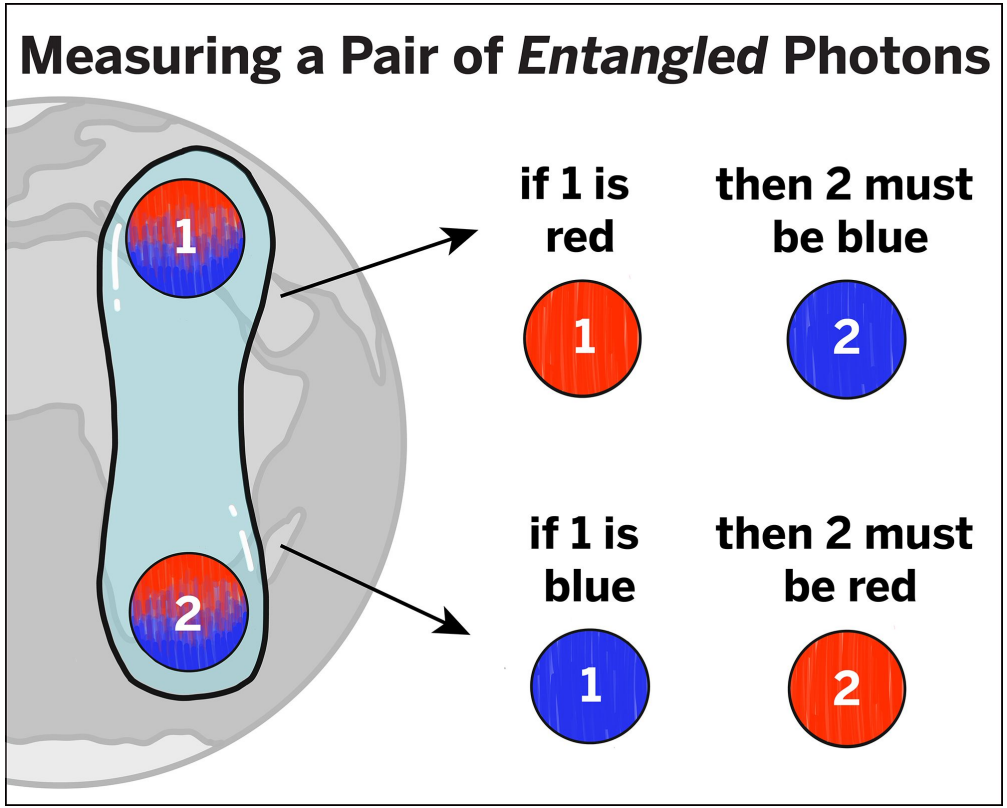
# Principle #2: Interference






*Double Slit Experiment*

# Principle #3: Entanglement



# … and that's how a quantum computer is built!



*David DiVincenzo*

How to build a quantum computer (DiVincenzo's Criteria)

1. Scalable system with 'nice' qubits
2. Be able to initialize your qubits to some known state
3. Qubits last a sufficiently long time
4. A 'universal' set of gates/operations
    - Superposition
    - Interference (via phases)
    - Entanglement
5. Ability to measure qubits

# What will you learn at qLearn?

Hardware (Michael)

- How to build a qubit
- Different hardware types
- Industrial quantum computers
- Quantum technologies at work (networks, sensing, etc)

Algorithms (Arjun)

- How to use a quantum computer
- Quantum information processing
- Algorithm classes
- Code-your-own quantum computer
- How to hack the government in $O((\log N)^2(\log\log N)(\log\log\log N))$ steps

# What’s going on at the UTQC?

REGISTER NOW!

Conference

qsiteconf.ca

Hackathon

qsitehacks.ca




Early bird sale!! $25 until **THIS FRIDAY**

# What's going on at the UTQC?

- **HACKATHONS!**
  - IonQ x Qollab Hackathon: Oct 9-11
  - HardHaQ: Mid-November

- **RESEARCH PROJECTS**
  - NV Diamond sensing project
  - Quantum PacBots challenge
  - **THIS FALL:** let us know!

- **OTHER**
  - **QLUIST:** present your research! (winter)

INDUSTRY PARTNERS & COLLABORATIONS



# What’s going on at the UTQC?

## Merch!









# Keep up-to-date!

utqc.sa.utoronto.ca

Discord Server

@uoft.quantum - Instagram

University of Toronto Quantum Computing Club - LinkedIn

If you're reading this, check out the qLearn syllabus under the 'Event's tab!